The Price of Fairness, Complexity, and Incentives in No-Numeraire Combinatorial Trade-Intent Auctions

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Abstract. Blockchain trade-intent auctions, which currently intermediate billions of dollars in trades per month, are combinatorial—a solver executing several intents jointly can net opposing flows and save on fees—yet they have no numeraire: there is no common asset in which to redistribute the resulting surplus. The deployed fix builds an endogenous fairness benchmark from the bids and executes a batched bid only if it delivers to every covered trader at least that trader's standalone competitive outcome. We give a worst-case theory of this design around one invariant: the complementarity factor g, together with liquidity $\theta = 1-\lambda$. The model and fairness benchmark are due to Canidio and Henneke [8]; our contribution is the worst-case approximation, competitive, and incentive layer on top.

The price of fairness is exactly $\min(g, 1/\theta)$, tight for all parameters and independent of the number of orders and of prices, with a dichotomy between g (no numeraire) and 1 (full numeraire); batch size is not the right parameter—the right refinement is a per-leg coverage floor α, under which the exact law is $\min(g, \alpha + (1-\alpha)/\theta)$. A Walrasian/UDP equilibrium supporting an efficient allocation exists for every instance iff $g = 1$, by native no-numeraire arguments, and at the trader-optimal equilibrium the price system is precisely the fairness benchmark. The price of strategyproofness is $\Theta(1+\log g)$ for a single multi-minded solver and for many single-minded solvers, via two posted-price mechanisms and a convex profit-curve technique; the remaining multi-agent multi-minded cell admits $O(N \log g)$ and a matching $\Omega(\min(N, g))$ barrier for serial-demand mechanisms. Online, the price splits: against realized lifetime floors even randomization pays exactly g, while with static public floors the residual problem is one-way search over the spread, deterministic $\Theta(\sqrt{g})$ and randomized $\Theta(\log g)$. Throughout, one scale-free family governs both the informational and the incentive price of the spread g.

Keywords: Combinatorial auctions · Price of fairness · Mechanism design · Blockchain · Competitive analysis.

1 Introduction

Blockchain "trade-intent" auctions let users state a desired exchange of one asset for another and delegate execution to competing agents called *solvers*; the protocol collects intents off-chain and auctions them in batches, in the spirit of frequent-batch market design [6]. These markets are economically significant—recent figures put monthly volume on the order of billions of dollars[1]—and they are *combinatorial*: a solver executing several intents together can net opposing flows and save on fees, producing more value than executing them separately; this batching logic is central in MEV-aware market design too [12,7]. What makes them unlike the combinatorial auctions of the literature is the absence of a *numeraire*. Users may demand any asset for any other, there is no common unit of account, and assets are illiquid, so the extra value created by batching cannot simply be paid out and shared.

The deployed response (the *fair combinatorial auction*, adopted by the studied protocol) constructs a fairness benchmark *from the bids themselves*: each trader's standalone competitive outcome defines a floor, and a batched bid is executed only if it delivers at least that floor to *every* covered trader. This sidesteps the impossibility of sharing batching gains by never allowing a batch to make any trader worse off than individual competition would. The equilibrium behavior of this design was analyzed by Canidio and Henneke [8], who identify a tension between the fairness guarantee and the value returned to traders. Thus the model and benchmark are theirs; our contribution is the worst-case approximation layer around them: how much welfare does the fairness filter cost, when do competitive equilibria exist, what does strategyproofness cost, and how do these prices behave online and under manipulation.

One invariant. We organize such a theory around a single structural quantity. For a solver with batch value $V_i(S)$ on a set S of intents and standalone values $V_i(\{j\})$, the *complementarity factor* is the least $g \geq 1$ with $V_i(S) \leq g \sum_{j \in S} V_i(\{j\})$ for all i, S: how much more a batch can be worth than the sum of its parts. The benchmark floor of order j is $c_j = \max_i V_i(\{j\})$, and liquidity is $\theta = 1 - \lambda \in [0, 1]$, where λ is the haircut on moving value across assets ($\theta = 0$ is the pure no-numeraire regime, $\theta = 1$ restores a numeraire). We show that g—together with θ—controls a remarkably wide range of worst-case quantities, which is what gives the design a unified analysis rather than a collection of unrelated bounds.

The role of g is deliberately worst-case: the theorems do not require a deployment to estimate the structural complementarity of every batch. The empirical section only shows that the corresponding observable spreads and supports are real in deployment.

Contributions. We ask what price three design requirements carry, each answered in terms of the same g (with liquidity θ). The main results are Theorems 1–3; the same spread parameter then governs incentives, online behavior, and benchmark manipulation.

[1] Order-of-magnitude figure from the deployment literature on the studied protocol; precise deployment identifiers are omitted for anonymity.

1. **Price of fairness (Section 3).** The welfare ratio between the unconstrained optimum and the best fair outcome is exactly $\min(g, 1/\theta)$, tight and independent of order count and of prices, with the no-numeraire/numeraire dichotomy $\{g, 1\}$. The binding instance is an asymmetry at the repair/drop indifference point; batch size is not the parameter, and under a per-leg coverage floor α the law is $\min(g, \alpha + (1-\alpha)/\theta)$.
2. **Equilibrium existence and the core (Section 4).** A Walrasian/UDP equilibrium supporting an efficient allocation exists for every instance iff $g = 1$, both directions proved natively; at the trader-optimal equilibrium the obligations are exactly the fairness floors. The deployed filter is precisely the singleton-coalition core constraint, the individual allocation always lies in the multiplicative g-core (tight), and the no-numeraire core can be nonempty even when the transferable core is empty.
3. **Price of strategyproofness (Section 5).** Two impossibilities pin down the regime: best-bid fairness conflicts with dominant-strategy truthfulness already on a single order, collapsing the enforceable benchmark to the runner-up level (Theorem 4), and online realized-floor manipulation forces ratio g even with randomization. Both make the endogenous benchmark strategically self-referential, so the minimal relaxation is a public, g-calibrated historical floor. In that exogenous-benchmark regime the price of strategyproofness is $\Theta(1+\log g)$—for many single-minded solvers and for one multi-minded solver—via an eligibility-posted packing mechanism and a coupled-menu mechanism with a convex profit-curve argument; independent option pricing pays $\Omega(g/\log^2 g)$, and the multi-agent multi-minded cell is pinned between $O(N \log g)$ and an $\Omega(\min(N, g))$ barrier.
4. **Online and strategic prices (Section 6).** Deterministic online fair settlement has competitive ratio exactly g; randomization does not help against realized lifetime floors, but with static public floors the residual problem is one-way search, deterministic $\Theta(\sqrt{g})$ and randomized $\Theta(\log g)$. The first-price game's individual restriction is smooth (coarse-correlated price of anarchy $\leq e/(e-1)$), and a contamination game prices benchmark manipulation with a sharp participation threshold and a rate-limit ramp tax.

Fair winner determination is NP-hard but fixed-parameter tractable in the number of directed asset pairs. Section 7 calibrates observable deployment diagnostics; the worst-case theory is independent of those measurements. Table 1 collects the quantitative results.

Related work. The model and the equilibrium analysis of the fair combinatorial auction are due to Canidio and Henneke [8]; we complement their 2×2 equilibrium study with a worst-case, approximation-theoretic analysis. Recent intent-market and solver-competition models give nearby strategic context [10], and batch trading has been proposed as a response to AMM arbitrage, LVR, and sandwich attacks [7]. Combinatorial auctions are classical [11], but almost always with a numeraire and quasilinear utilities; our setting has neither, putting it closer to mechanism design without money [17] and no-transfer impossibility

Table 1. Worst-case prices as functions of the complementarity factor g and liquidity θ (N: number of strategic solvers; α: coverage). "exact" marks literal equalities or fully tight characterizations; "asymptotically tight" marks $\Theta(\cdot)$ bounds.

Quantity	Result	Status
Price of fairness	$\min(g, 1/\theta)$; endpoints $\{g, 1\}$	exact
coverage refinement	$\min(g, \alpha + (1-\alpha)/\theta)$; size-irrelevant	exact
Walrasian/UDP existence	for all instances iff $g = 1$	exact
Core	filter = singleton-core; g-core nonempty, tight	exact
Winner determination	NP-hard; FPT in directed pairs P	—
Strategyproofness (screening)	$\Theta(1 + \log g)$; det. exactly $\sqrt{g}$	asympt. tight
single-/multi-minded	both $\Theta(1 + \log g)$; indep. $\Omega(g/\log^2 g)$	asympt. tight
multi-agent multi-minded	$O(N \log g)$ / serial $\Omega(\min(N, g))$	partial (serial barrier)
Online (realized floors)	exactly g, even randomized	exact
Online (static floors)	det. $\Theta(\sqrt{g})$, rand. $\Theta(\log g)$	asympt. tight
Price of anarchy	individual-only $\leq e/(e-1)$; full conj. $\Theta(g)$	partial (latent batches)
Contamination	threshold $\kappa_1 \geq v\beta$ deters; $g \mapsto g/(1-\mu)$	—

phenomena in MEV-aware fee mechanisms [1]. The price of fairness was introduced for divisible resources [4] and developed for indivisible goods under proportionality, envy-freeness, and MMS [9,3,2]; here it scales with production complementarity and liquidity, not agent count. Equilibrium existence draws on assignment markets [19] and package assignment [5]. Our smoothness bound uses Syrgkanis–Tardos [20] and intrinsic robustness [18]; online search instantiates one-way trading [13]. Order-flow-auction models under PBS provide related strategic context [16].

2 Model

There are n trade intents (*orders*) and a set of *solvers*. A solver i has private *deliverable values* $v_{i,j} \geq 0$ for orders: the best outcome it can produce for trader j, in j's requested asset and normalized score units. Its batch value is $V_i(S) = \sum_{j \in S} v_{i,j}$, realized only if the batch executes. A winning bid delivers $w_j \leq v_{i,j}$ to each covered order; the solver's profit is the retained value $V_i(S) - \sum_{j \in S} w_j$, kept in the produced assets rather than transferred across traders. Reports are *binding*: a solver cannot claim value it cannot deliver, so the strategic misreports relevant for our posted models are downward. Welfare $W = \sum_j w_j$ is an additive social objective over trader-normalized delivered scores, not a transferable monetary surplus.

Definition 1 (Complementarity factor). *The instance has complementarity factor $g \geq 1$ if $V_i(S) \leq g \sum_{j \in S} V_i(\{j\})$ for every solver i and every S. Thus g caps how much batching amplifies standalone value; $g = 1$ is additive.*

Definition 2 (Fairness floor; fair allocation). *The floor of order j is $c_j := \max_i V_i(\{j\})$, the value of its best standalone bid; $C := \sum_j c_j$. An allocation is fair if every allocated order receives $w_j \geq c_j$; batched bids violating this for some covered order are filtered.*

An allocation selects winning bids covering disjoint order sets; its welfare is $W = \sum_j w_j$. Let OPT be the maximum welfare ignoring fairness and FAIR the maximum over fair allocations; the *price of fairness* is $\mathrm{PoF} := \sup \mathrm{OPT}/\mathrm{FAIR}$ over a stated class.

Definition 3 (Liquidity). *Within a single winning bid a solver may convert value across its traders at haircut $\lambda \in [0, 1]$: withdrawing x from one trader's asset yields $(1-\lambda)x$ for another. Write $\theta := 1 - \lambda$ for liquidity; $\theta = 0$ is the pure no-numeraire case, $\theta = 1$ a full numeraire. Conversion is per-bid.*

Per-bid conversion models a solver rebalancing net flows *within* a single settlement (its own routing), not across settlements or traders; this is a stylized but conservative abstraction of deployed solver behavior.

The deployed mechanism additionally requires uniform clearing prices per directed token pair (*uniform directed prices*, UDP), forcing all orders on one directed pair into a single winning bid. UDP is vacuous in Sections 3–5 (our constructions use distinct pairs) and reappears as the source of fixed-parameter tractability. This is the order-level analogue of the uniform clearing price in frequent batch auctions [6], specialized to the trade-intent model of Canidio and Henneke [8]. Full proofs of all numbered results are in the appendix.

Running example. The basic tension already appears with two orders. Suppose order 2 has floor c and order 1 has essentially no standalone floor, while a batch creates total value γc mostly on order 1. Fairness forces the batch either to transfer value across assets to repair order 2, or to drop the batch and settle floors. This repair/drop tradeoff is the geometry behind Theorem 1; the sharp loss occurs when $\gamma = \min(g, 1/\theta)$.

3 The Price of Fairness

The first price is informational: once the benchmark floors are fixed by standalone competition, how much welfare can the componentwise fairness filter destroy? The answer is governed exactly by g and liquidity θ.

Theorem 1 (Price of fairness, exact). *For every $\theta \in [0, 1]$ and $g \geq 1$, over all instances with liquidity θ and complementarity factor at most g,*

$$\mathrm{PoF}(\theta, g) = \min\big(g,\, 1/\theta\big),$$

tight, and independent of the number of orders and of prices ($1/\theta := \infty$ at $\theta = 0$).

Proof (Proof idea). Upper bound. Process each bid of an unconstrained optimum independently. A bid with gain $\gamma = V/C_S \in [1, g]$ has deficit $D = \sum_{j\in S}(c_j - v_j)^+ \leq C_S$. Either drop the batch and settle floors, delivering $C_S = V/\gamma$, or repair starved legs using the batch's own surplus. Clearing D loses $D(1-\theta)/\theta$, so whenever repair is feasible,

$$\frac{V - D(1-\theta)/\theta}{V} \geq 1 - \frac{1-\theta}{\theta\gamma} \geq \theta.$$

Dropping covers low gains by V/g; repair covers high gains by $V/(1/\theta)$; summing over disjoint optimal bids gives the bound. *Lower bound.* A two-order instance with one starved leg, calibrated to $\gamma^* = \min(g, 1/\theta)$, forces FAIR $= c$ against OPT $= \gamma^* c$. The binding instance is an extreme asymmetry, not maximal complementarity. □

Corollary 1 (Dichotomy). *At the endpoints, the pure no-numeraire auction pays the full factor g, while a full numeraire removes the fairness cost entirely.*

A natural attempt to add a third parameter fails, and identifies the right one.

Proposition 1 (Batch size is not the parameter). *With* $\mathrm{PoF}(\theta, g, d)$ *the price restricted to bids covering at most d orders,*

$$\mathrm{PoF}(\theta, g, 1) = 1, \qquad \mathrm{PoF}(\theta, g, d) = \min(g, 1/\theta) \quad (d \geq 2).$$

Say an instance has *coverage* $\alpha \in [0, 1]$ if every bid satisfies $v_j \geq \alpha c_j$ on every covered order—no leg is worse than α times the benchmark, a per-leg routing-quality floor.

Theorem 2 (The coverage law). *Write $\Gamma(\theta, \alpha) := \alpha + (1-\alpha)/\theta$, with $\Gamma(0, \alpha) = \infty$ for $\alpha < 1$ and $\Gamma(0, 1) = 1$. Over instances with liquidity θ, complementarity at most g, and coverage α,*

$$\mathrm{PoF}(\theta, g, \alpha) = \min\bigl(g, \Gamma(\theta, \alpha)\bigr),$$

tight on two-order instances.

The quantity $\Gamma - 1 = (1-\alpha) \cdot \frac{1-\theta}{\theta}$ is the largest relative per-leg shortfall times the conversion loss per unit of delivered deficit: the price of fairness prices the worst starved leg, discounted by liquidity. The endpoint $\alpha = 1$ is exactly the componentwise filter-passing under which fairness is free. This refinement is also why batch size disappears from the theorem: what matters is not how many orders are bundled, but how much one covered leg can fall below its own standalone benchmark before the batch's surplus can be converted back to repair it.

4 Equilibrium Existence and the Core

The second question is structural. When does the same filter coincide with a competitive equilibrium or a cooperative stability notion rather than merely a feasibility rule? Here $g = 1$ is the threshold. This is not a restatement of the gross-substitutes existence theory: gross substitutes support Walrasian equilibria with prices and transfers over items [15,14], whereas our frontier is about componentwise obligations in traders' requested assets.

We work in the order-price abstraction of UDP: a directed-rate system induces order-level obligations $w_j \geq 0$ that any solver serving j must deliver, and the single property used is that w_j depends only on the order. A *UDP equilibrium supporting the efficient allocation* is a rate system and an allocation in which every order is served at exactly its w_j, every solver's executed portfolio maximizes its profit $\sum_j (v_j - w_j)$ over feasible (executable, i.e. $v_j \geq w_j$) portfolios, and produced value equals OPT. This definition deliberately separates prices from payments: w_j is an obligation in order j's normalized delivered units, not a transfer that can compensate another trader. That separation is what makes the frontier different from the transferable package-assignment theorem.

Lemma 1 (Batch domination under $g = 1$). *If $g = 1$ then at any rates every executable batched bid is weakly dominated by the sub-portfolio of its positive-profit singletons: $\sum_{j \in S}(v_j - w_j) = V_i(S) - \sum_j w_j \leq \sum_j (V_i(\{j\}) - w_j)^+$. Hence every solver has an individual-only best response; the one use of UDP is that w_j is the same in a batch or a singleton.*

The lemma is the reason $g = 1$ is special. If batching is never more productive than the sum of singletons, then any profitable executable batch contains enough profitable singleton mass to replace it. The no-numeraire feasibility constraint only strengthens this conclusion: some batch deviations that would be available with transfers are simply not executable componentwise. Conversely, once $g > 1$, overlap matters. A solver can create value on a pair without creating enough value on either order alone to support order-level obligations. The three-cycle lower bound exploits exactly this gap: each pair wants one obligation low for feasibility and another high for incentive compatibility, and the cycle leaves no price vector that satisfies all three demands.

Theorem 3 (Existence frontier is $g = 1$). *A UDP equilibrium supporting an efficient allocation exists for every instance of the class with complementarity at most g iff $g = 1$. For $g = 1$ the obligations $w_j = c_j$ are an equilibrium, so the equilibrium is best-bid fair; for every $g > 1$ some instance admits none. (Solvers have no capacity constraints; in (a) the floor vector is assumed rate-realizable.)*

Proof (Proof idea). $g = 1$: set $w_j = c_j$ and assign each order to a best producer; winners earn zero margin, losers nonpositive, and Lemma 1 kills batch deviations. Produced value is $C = \mathrm{OPT}$, and the price vector coincides with the fairness benchmark. *$g > 1$:* the obstruction is native—non-existence with transfers does not imply it here, since feasibility only *removes* deviations. The hard instance is a three-cycle of overlapping pair-batches with vector $(1+\frac{\delta}{2}, 1+\frac{\delta}{2})$ over floors 1. Supporting one efficient pair forces an obligation on the third order low enough to keep that pair feasible; the two alternative pairs then impose deviation inequalities that sum to $\delta \leq 0$, contradicting any positive complementarity. Full details are in Appendix B. □

The filter also has a cooperative reading. A bid *blocks* an allocation at factor κ if some feasible delivery gives every covered order strictly more than κw_j; the κ-core forbids this.

Proposition 2 (Core placement). *(i) Best-bid fairness is exactly immunity to singleton blocking at factor* 1*: the deployed filter is the singleton-coalition core constraint. (ii) The individual allocation $w_j = c_j$ lies in the g-core for every θ, and g is the least universal blocking-immunity factor. (iii) There are instances whose transferable-utility core is empty while the $\theta = 0$ core is nonempty—a separation driven by overlap, like the integrality gap.*

Thus the fairness filter is not an ad hoc admissibility test; it is the smallest coalition-stability condition visible when coalitions are restricted to orders' own requested assets. The multiplicative g-core statement says that even when exact support fails, the individual allocation is never blocked by more than the same complementarity factor that drives Section 3. This is one of three appearances of a single phenomenon: the no-numeraire constraint *removes deviations* (VCG payments become infeasible; batch deviations are dominated at $g = 1$; componentwise blocking is weak). Whether the $\theta = 0$ core is always nonempty is open: Scarf's route needs balancedness, broken by the same overlap geometry.

Complexity. Fair winner determination is NP-hard (from 3-dimensional matching) but fixed-parameter tractable in the number P of directed asset pairs: UDP contracts each directed pair to an atomic item, and every fair bid is a weighted subset of the P atoms. Dynamic programming over pair masks (or equivalently weighted set packing on P items) tries 2^P states and scans the bid list in each state, giving $2^{O(P)} \cdot \mathrm{poly}(|\mathcal{B}|)$. The deployment cap on pairs per batch thus makes the problem tractable in exactly the deployed regime. This is another way in which g and the token-pair graph play complementary roles: g controls the loss from fairness and incentives, while P controls the search space left by the uniform-price constraint.

5 The Price of Strategyproofness

The third price is strategic. Truthfulness is not studied under the original endogenous benchmark, because the preceding structure creates a trilemma: current-report floors, ex-post fairness, and VCG-style strategyproofness cannot coexist without transfers. This places the problem in the no-money mechanism-design tradition [17], with a blockchain-specific impossibility flavor close to post-MEV fee mechanism design [1].

Two impossibilities pin down the regime. First, with *endogenous* floors—computed from the current reports—VCG-style compensation is infeasible because the no-numeraire constraint gives no asset in which to make the Clarke payment. More sharply, best-bid fairness and dominant strategy truthfulness already conflict on one order:

Theorem 4 (Best-bid fairness is incompatible with strategyproofness). *Call a mechanism non-wasteful if it allocates an order whenever some reported standalone value is positive, and value-respecting if it allocates to a solver with maximal reported value. No feasible, non-wasteful, value-respecting, DSIC mechanism guarantees fairness at the endogenous best-bid benchmark $c_j = \max_i r_{ij}$. This holds already for one order. Under DSIC, the enforceable benchmark collapses to the runner-up level.*

Second, as Section 6 proves, online with realized lifetime floors admits no fair mechanism better than ratio g, even randomized: a solver settling a batch can be undercut by a later standalone bid that raises a covered order's own floor. Both failures trace to the same source—when the floor responds to current behavior, a solver can manipulate its own benchmark—so the natural and *deployed* response is to fix floors from a public, g-calibrated historical window, severing that channel while preserving the spread g as the operative parameter. We therefore analyze this exogenous-benchmark regime; the price of strategyproofness PoSP is the welfare ratio between FAIR and the best universally truthful fair mechanism, and solvers are single- or multi-minded with private values passing the filter. The endogenous results above show this is a relaxation forced by impossibility, not a convenience: under endogenous floors the relevant question is not the price of strategyproofness but its non-existence.

This reduction is important for interpretation. The public-floor model is not a stronger assumption smuggled in to obtain mechanisms; it is the minimal relaxation left after the endogenous benchmark has been shown to be strategically self-referential. The theorem below therefore measures the residual cost of truthfulness once that self-reference has been severed.

The canonical screening family of the regime is already nontrivial for one strategic solver:

Theorem 5 (Canonical screening family). *Over the screening family R_g (one strategic solver, public floors), the price of strategyproofness is $\Theta(1+\log g)$: every randomized DSIC mechanism pays at least $\frac{g \ln g}{2(g-1)} \geq \frac{\ln g}{2}$, a geometric posted-price mechanism achieves $e(\lceil \ln g \rceil + 1)$, and the best deterministic mechanism is exactly $\sqrt{g}$.*

The lower bound is a model-native technique: a one-sided envelope inequality (downward incentive compatibility alone forces monotone utilities, so the utility derivative dominates the execution probability a.e.) integrated against a V^{-2} kernel. In the screening family a type V can only underreport; if $x(V)$ is execution probability and $u(V)$ interim utility, downward IC gives

$$u(V) \geq u(V') + (V - V')x(V') \qquad (V' \leq V),$$

so $u'(V) \geq x(V)$ a.e.; the scale-free integral forces logarithmic loss. The deterministic optimum balances the two errors at threshold $\sqrt{g}$. Random geometric posted levels give the matching upper bound up to constants. Two mechanisms lift the guarantee to the full game.

Mechanism model. In the exogenous-benchmark regime the floors c_j are public. A solver's feasible options and their floor masses $C_S = \sum_{j \in S} c_j$ are public; only its values V_S are private. The mechanisms below are posted: prices and eligibility depend only on public data and a public random seed, and each solver responds by demanding a disjoint collection of options rather than reporting values. The mechanism honors demand only when the solver delivers the floors to every covered order. Universal truthfulness is in this posted-demand sense: for every seed, demanding a utility-maximizing executable collection is dominant because the menu is independent of that solver's behavior.

Theorem 6 (Two routes to $\Theta(1 + \log g)$). *(i) For arbitrarily many single-minded solvers with public floors, an eligibility-posted packing mechanism is universally truthful and $\Theta(1 + \log g)$-competitive. (ii) For one multi-minded solver of arbitrary structure, a coupled-menu mechanism—post all options at prices $C_S\, t$ for a single random multiplier $t = e^k$ and let the solver demand any profit-maximizing disjoint collection—is universally truthful and $\Theta(1 + \log g)$-competitive. Independent per-option pricing instead pays $\Omega(g/\log^2 g)$.*

The multi-minded bound rests on a *profit-curve* technique: the solver's profit $p(t)$ over disjoint collections is convex and piecewise linear, delivered mass is $t\,(-p'(t))$, and a geometric grid telescopes $p(1)$ against expected welfare. The single random multiplier is essential: independently pricing options lets a multi-minded solver take only the cheapest member of a family of substitutes, losing the collection value that fairness must compare against. The remaining cell is both multi-agent and multi-minded:

Theorem 7 (Multi-agent multi-minded). *Random-priority coupled menus are universally truthful with $\mathbb{E}[W] \geq \mathrm{FAIR}/\big(N(e-1)(\lceil \ln g \rceil + 1) + 1\big)$, i.e. $\Theta_N(1 + \log g)$ for a constant number N of strategic solvers. Conversely every serial-demand mechanism—a report-independent ex-ante service order, arbitrary per-identity multiplier distributions, each agent demanding a profit-maximizing collection—pays $\Omega(\min(N, g))$, even with geometry-aware ordering. So any N-independent* $\mathrm{polylog}(g)$ *mechanism must break the serial-demand format.*

The barrier is proved by a *symmetrized swarm*: all identities share the same public geometry (all singletons plus the grand set), differing only in private values, so no report-independent order can tell the batcher from the blockers. The telescoping identity

$$\sum_i (1 - q_i) \prod_{s \prec i} q_s = 1 - \prod_s q_s \leq 1$$

then places the batcher behind an expected blocker.

These results should be read as a trichotomy. Single-minded competition is handled by posted eligibility and packing; one multi-minded solver is handled by a coupled menu and profit-curve telescoping; many multi-minded solvers expose a genuine serial-demand barrier. The open question is whether a mechanism that is not serial-demand can remove the dependence on N while keeping the logarithmic dependence on g.

6 Online and Robustness

The remaining results explain why deployed systems replace current-report floors by historical, public, and rate-limited ones.

Online floors. With realized lifetime floors, online fairness can be attacked by a later standalone bid that raises a covered order's own floor. This is the online analogue of the endogenous-benchmark impossibility in Section 5: the solver competes against a future revision of the fairness constraint itself.

Proposition 3 (Realized floors force a factor g). *The deterministic online competitive ratio of fair settlement is exactly g.*

Theorem 8 (Two online prices). *(i) Against realized lifetime floors with almost-sure fairness, randomization does not help: the competitive ratio is exactly g. (ii) With static public floors on single-contention instances, the residual problem is one-way search over the spread, with deterministic ratio $\Theta(\sqrt{g})$ and randomized ratio $\Theta(\log g)$; ratio-threshold greedy mechanisms still pay $\Omega(g)$.*

Thus Theorem 8 is the operational reason for the exogenous-benchmark regime in Section 5; once floors are fixed, the residual problem is one-way search over width g [13].

Strategic play. For the first-price game, measured in produced welfare PW rather than delivered welfare, the individual-only restriction is smooth in the standard composable-mechanisms sense [20,18]:

Theorem 9 (Individual competition is smooth). *Every coarse correlated equilibrium of the per-order-decomposable individual-only game has* $\mathbb{E}[\mathrm{PW}] \geq (1 - \frac{1}{e})\mathrm{OPT}_{\mathrm{ind}}$*, hence price of anarchy at most $e/(e-1)$ relative to* $\mathrm{OPT}_{\mathrm{ind}}$ *and at most $\frac{e}{e-1}\,g$ relative to the unrestricted optimum. The full-game price of anarchy is conjectured $\Theta(g)$; the obstruction is latent batches that execute only under a deviation.*

Benchmark robustness. On one directed pair, let an attacker of volume v depress a quantile benchmark by injecting mass a at convex cost $K(a) = \kappa_1 a + \frac{\kappa_2}{2}a^2$, with local sensitivity β.

Theorem 10 (The contamination game). *A marginal injection cost $\kappa_1 \geq v\beta$ deters contamination. Otherwise the optimal steady attack is $a^* = \min\{\bar{a}, (v\beta - \kappa_1)/\kappa_2\}$, inducing relative bias $\mu = \beta a^*/b^*$; bursts do not beat the constant attack, and rate limits tax restarts through a finite ramp without changing the steady state. Every floor-calibrated guarantee is then read with $g \mapsto g/(1-\mu)$.*

Fees deter, detection tempers, and rate limits tax ramping; persistent bias is removed only by a sufficient cost floor.

7 Empirical Calibration

An anonymized supplementary deployment trace contains 802,816 post-adoption solver competitions over eight months. On the sane-volume USD subsample, the trace reports \$54.69B of covered notional and a \$5.13M visible accounting gap, about 0.94 bps. The visible spread is nontrivial in the tail: $\widehat{G}^{\mathrm{obs}} > 2$ in 25.3% of auctions, $\widehat{G}^{\mathrm{obs}} > 4$ in 20.2%, and $\widehat{G}^{\mathrm{obs}} > 2$ in 63.1% of small-notional auctions. The supplement defines the score variables, filters, and replay diagnostics. The point is not to estimate structural g. The bounds are design guardrails: they say what happens if flow drifts toward high-complementarity or thin-liquidity assets, exactly where the small-notional tail is largest. They also remain relevant under manipulation, since benchmark contamination feeds back as $g \mapsto g/(1-\mu)$. The same trace makes P observable and motivates benchmark robustness: under a naive local hard-reserve benchmark, 97.1% of auction-weighted cell-windows are vulnerable, a design signal for inherited, quantile-based, delayed, and rate-limited benchmarks.

These are observable deployment diagnostics, not estimates of the structural complementarity factor g; the theory's worst-case bounds hold for whatever g a deployment exhibits.

8 Conclusion

A single invariant—the complementarity factor g, with liquidity θ—governs the worst-case fairness, equilibrium, complexity, incentive, online, and manipulation prices of the no-numeraire fair combinatorial auction. Two phenomena recur: no-numeraire feasibility removes some deviations while disabling transfers, and one geometric family prices both informational and incentive costs of the spread.

Two obstructions remain deliberately visible. The *latent-batch* obstruction blocks full-game smoothness because deviations can activate batches absent under the original profile. The *serial-demand barrier* shows that report-independent service orders can lose $\Omega(\min(N, g))$. Removing either seems to require a new idea.

References

1. Bahrani, M., Garimidi, P., Roughgarden, T.: Transaction fee mechanism design in a post-MEV world. In: 6th Conference on Advances in Financial Technologies

(AFT 2024). Leibniz International Proceedings in Informatics (LIPIcs), vol. 316, pp. 29:1–29:24. Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024). https://doi.org/10.4230/LIPIcs.AFT.2024.29

2. Barman, S., Bhaskar, U., Shah, N.: Optimal bounds on the price of fairness for indivisible goods. In: Web and Internet Economics (WINE). LNCS, vol. 12495, pp. 356–369. Springer (2020)
3. Bei, X., Lu, X., Manurangsi, P., Suksompong, W.: The price of fairness for indivisible goods. In: Proc. 28th International Joint Conference on Artificial Intelligence (IJCAI). pp. 81–87 (2019)
4. Bertsimas, D., Farias, V.F., Trichakis, N.: The price of fairness. Operations Research **59**(1), 17–31 (2011)
5. Bikhchandani, S., Ostroy, J.M.: The package assignment model. Journal of Economic Theory **107**(2), 377–406 (2002)
6. Budish, E., Cramton, P., Shim, J.: The high-frequency trading arms race: Frequent batch auctions as a market design response. The Quarterly Journal of Economics **130**(4), 1547–1621 (2015). https://doi.org/10.1093/qje/qjv027
7. Canidio, A., Fritsch, R.: Arbitrageurs' profits, LVR, and sandwich attacks: Batch trading as an AMM design response (2023), arXiv:2307.02074
8. Canidio, A., Henneke, F.: Fair combinatorial auction for blockchain trade intents: Being fair without knowing what is fair (2024), arXiv:2408.12225
9. Caragiannis, I., Kaklamanis, C., Kanellopoulos, P., Kyropoulou, M.: The efficiency of fair division. Theory of Computing Systems **50**(4), 589–610 (2012)
10. Chitra, T., Kulkarni, K., Pai, M., Diamandis, T.: An analysis of intent-based markets (2024), arXiv:2403.02525
11. Cramton, P., Shoham, Y., Steinberg, R. (eds.): Combinatorial Auctions. MIT Press (2006)
12. Daian, P., Goldfeder, S., Kell, T., Li, Y., Zhao, X., Bentov, I., Breidenbach, L., Juels, A.: Flash boys 2.0: Frontrunning, transaction reordering, and consensus instability in decentralized exchanges. In: 2020 IEEE Symposium on Security and Privacy (SP). pp. 910–927. IEEE (2020)
13. El-Yaniv, R., Fiat, A., Karp, R.M., Turpin, G.: Optimal search and one-way trading online algorithms. Algorithmica **30**(1), 101–139 (2001)
14. Gul, F., Stacchetti, E.: Walrasian equilibrium with gross substitutes. Journal of Economic Theory **87**(1), 95–124 (1999)
15. Kelso, A.S., Crawford, V.P.: Job matching, coalition formation, and gross substitutes. Econometrica **50**(6), 1483–1504 (1982)
16. Ma, R., Tang, W., Yao, D.: Analysis of the order flow auction under proposer-builder separation on blockchain (2025), arXiv:2502.12026
17. Procaccia, A.D., Tennenholtz, M.: Approximate mechanism design without money. ACM Transactions on Economics and Computation **1**(4) (2013), article 18; preliminary version in EC 2009, pp. 177–186
18. Roughgarden, T.: Intrinsic robustness of the price of anarchy. Journal of the ACM **62**(5), 32:1–32:42 (2015)
19. Shapley, L.S., Shubik, M.: The assignment game i: The core. International Journal of Game Theory **1**(1), 111–130 (1971)
20. Syrgkanis, V., Tardos, É.: Composable and efficient mechanisms. In: Proc. 45th ACM Symposium on Theory of Computing (STOC). pp. 211–220 (2013)

A Full Proofs for Section 3

Proof (Proof of Theorem 1). Upper bound. Fix an unconstrained optimum and process its winning bids independently; bids cover disjoint order sets, so the repaired welfares add. Take a winning bid with floor mass $C_S = \sum_{j\in S} c_j > 0$ (a bid with $C_S = 0$ has all floors zero and is already fair), value $V = \sum_{j\in S} v_j$, gain $\gamma = V/C_S \in [1, g]$, and deficit $D = \sum_{j\in S}(c_j - v_j)^+$. Two fair uses:

Repair. Move value from over-served legs to starved ones. The total surplus available is $s = V - (C_S - D) = V - C_S + D$, and clearing the deficit costs D/θ units of surplus (haircut $\lambda = 1-\theta$). This is feasible iff $V - C_S \geq D\frac{1-\theta}{\theta}$, and then the delivered welfare is $f = V - D\frac{1-\theta}{\theta}$. Since every delivered value is nonnegative, $(c_j - v_j)^+ \leq c_j$ and therefore $D \leq C_S$. Together with $V - C_S = (\gamma - 1)C_S$, feasibility holds whenever $\gamma - 1 \geq \frac{1-\theta}{\theta}$, i.e. $\gamma \geq 1/\theta$; then

$$\frac{f}{V} = 1 - \frac{1-\theta}{\theta\gamma}\frac{D}{C_S} \geq 1 - \frac{1-\theta}{\theta\gamma} \geq \theta = \frac{1}{1/\theta}.$$

Thus repair gives $f \geq V/(1/\theta)$ whenever $\gamma \geq 1/\theta$, and hence $f \geq V/\min(g, 1/\theta)$ in the repair regime.

Drop. Settle every covered order at its floor: $f = C_S = V/\gamma$. For $\gamma \leq \min(g, 1/\theta)$ this is $\geq V/\min(g, 1/\theta)$.

Either way each bid retains a $1/\min(g, 1/\theta)$ fraction of its value; summing, $\text{FAIR} \geq \text{OPT}/\min(g, 1/\theta)$.

Lower bound. Let $\gamma^* = \min(g, 1/\theta)$. Two orders on distinct pairs, $c_1 = 0$, $c_2 = c$; one batched bid with $v_2 = 0$, $v_1 = \gamma^* c$, so $V = \gamma^* c$, $C_S = c$, and the batching solver's standalone values are $0, c$, so Definition 1 holds as $\gamma^* \leq g$. Repairing order 2 to its floor c requires withdrawing c/θ from v_1; if $\gamma^* = 1/\theta$ this exhausts v_1 exactly (welfare c), and if $\gamma^* = g < 1/\theta$ the batch is irreparable (welfare c from the individual allocation). In both cases $\text{FAIR} = c$ and $\text{OPT} = \gamma^* c$, so $\text{OPT}/\text{FAIR} = \gamma^*$. Independence of the number of orders and of prices is immediate from the construction. □

Proof (Proof of Proposition 1). The two-order family of Theorem 1 already has $d = 2$ and attains $\min(g, 1/\theta)$, and the upper bound never uses d; for $d = 1$ only singletons exist, so $\text{OPT} = \text{FAIR} = C$. □

Proof (Proof of Theorem 2). Upper bound. Write $\Gamma = \Gamma(\theta, \alpha) := \alpha + (1-\alpha)/\theta$, with the endpoint convention from the theorem. As above, coverage caps the deficit by $D \leq (1-\alpha)C_S$. Repair is feasible when $\gamma - 1 \geq (1-\alpha)\frac{1-\theta}{\theta} = \Gamma - 1$, and then $f = V - \frac{D(1-\theta)}{\theta} \geq V\big(1 - \frac{(1-\alpha)(1-\theta)}{\theta\gamma}\big) \geq V/\Gamma$ for $\gamma \geq \Gamma$ (monotone in γ, equality at $\gamma = \Gamma$); dropping gives $V/\gamma \geq V/\Gamma$ for $\gamma \leq \Gamma$ and V/g for $\gamma \leq g$. Hence $\text{FAIR} \geq \text{OPT}/\min(g, \Gamma)$. *Lower bound.* Two orders, $c_1 = 0$, $c_2 = c$; batched bid with $v_2 = \alpha c$ (coverage saturated) and $v_1 = (\gamma^* - \alpha)c$, $\gamma^* = \min(g, \Gamma)$. Repairing order 2 needs $(1-\alpha)c/\theta$ from v_1; if $\gamma^* = \Gamma$ then $v_1 = (\Gamma - \alpha)c = \frac{(1-\alpha)c}{\theta}$ exactly, and if $\gamma^* = g < \Gamma$ the batch is irreparable. Either way $\text{FAIR} = c$, $\text{OPT} = \gamma^* c$. □

B Full Proofs for Section 4

Proof (Proof of Lemma 1). $\sum_{j\in S}(v_j - w_j) = V_i(S) - \sum_{j\in S} w_j \le \sum_{j\in S} V_i(\{j\}) - \sum_{j\in S} w_j \le \sum_{j\in S}(V_i(\{j\}) - w_j)^+$, the middle step by $g = 1$. Each positive-profit singleton has $V_i(\{j\}) > w_j$, hence is executable; a non-executable batch is not a deviation. The obligation w_j is the same whether j is served in the batch or individually, which is the only property of UDP used. □

Proof (Proof of Theorem 3). (a), $g = 1$. Set $w_j = c_j$ and assign each order j to a solver attaining $V_i(\{j\}) = c_j$. The winner earns zero margin per order; any losing singleton has $V_i(\{j\}) - c_j \le 0$; and Lemma 1 kills batch deviations (no positive-profit singletons). With $g = 1$, any allocation produces at most $\sum_{(i,S)} V_i(S) \le \sum_j c_j = C$, attained here, so produced value is $C = \mathrm{OPT}$, all delivered. The point $w_j = c_j$ is the best-bid floor, so the equilibrium is fair (the trader-optimal core point of the underlying assignment market [19], to which Lemma 1 reduces the economy).

(b), $g > 1$. We argue natively; non-existence with transfers does not transfer, as feasibility only removes deviations. Fix $\delta \in (0, 2(g-1)]$; orders $1, 2, 3$ on distinct pairs, a rival with standalone value 1 on each ($c_j = 1$), and three solvers each holding one batched bid on a pair $\{j, j'\}$ with vector $(1 + \frac{\delta}{2}, 1 + \frac{\delta}{2})$ (value $2 + \delta \le 2g$) and standalone values 1. The efficient allocation runs one batch, say $\{1, 2\}$, plus order 3 individually (produced $3 + \delta$). Suppose rates w support it. Order 3 is served by a standalone solver of value 1, so $w_3 \le 1$; orders $1, 2$ are served by the batch, so $w_1, w_2 \le 1 + \frac{\delta}{2}$. The batch $\{2, 3\}$ is then executable, and equilibrium requires it unprofitable against its solver's individual alternatives: $(2 + \delta) - w_2 - w_3 \le (1 - w_2)^+ + (1 - w_3)^+$. If $w_2 \le 1$ the right side is $(1 - w_2) + (1 - w_3)$, giving $\delta \le 0$; if $w_2 > 1$ it is $1 - w_3$, giving $w_2 \ge 1 + \delta > 1 + \frac{\delta}{2}$, contradicting feasibility. No supporting rates exist. □

Proof (Proof of Proposition 2). (i) If $w_j < c_j$, the standalone bid attaining c_j blocks j at factor 1; if $w_j \ge c_j$ for all j, no singleton strictly improves. (ii) Blocking $w_j = c_j$ at factor g needs delivery $> g\, c_j$ to every $j \in S$, totalling $> g\, C_S$, while any feasible delivery from a bid totals $\le V_i(S) \le g\, C_S$ (haircuts only lose value; if all floors in S are zero then $V_i(S) = 0$ and strict blocking is impossible). Tightness: a batch with vector $(g'c, g'c)$ over floors (c, c), $g' \in (\kappa, g]$, blocks at any $\kappa < g$. (iii) Three-cycle floors $(1, 1, 1)$, pair vectors $(1 + \frac{\delta}{2}, 1 + \frac{\delta}{2})$: summing the three transferable core constraints $x_j + x_{j'} \ge 2 + \delta$ gives $\sum x_j \ge 3 + \frac{3\delta}{2} > 3 + \delta = \mathrm{OPT}$, so the transferable core is empty; the allocation (batch $\{1, 2\}$, order 3 individually) is $\theta = 0$-unblocked since every other batch caps a covered order at a value already attained. □

Complexity. NP-hardness is a reduction from 3-dimensional matching: orders are ground elements, batched bids are triples of value 1, and a fair welfare-n allocation exists iff a perfect matching does. For fixed-parameter tractability, UDP contracts each directed pair to an atomic item. After fair filtering, every surviving bid is a weighted subset of the P atoms, and winner determination is weighted set packing on a universe of size P. The dynamic program $DP[M] = \max\{DP[M \setminus T_b] + w_b : b \text{ uses } T_b \subseteq M\}$ over masks $M \subseteq [P]$ scans the bid list at each state, giving $2^{O(P)} \cdot \mathrm{poly}(|\mathcal{B}|)$.

C Full Proofs for Section 5

Proof (Proof of Theorem 4). Let there be one order. Solver i has private deliverable value v_i and may report any binding value $r_i \le v_i$; over-reporting is excluded by settlement, while under-reporting is feasible. By non-wastefulness the order is allocated whenever some report is positive, and by value-respecting the winner has a maximal report. Feasibility implies the winner can deliver at most its true value. Best-bid fairness requires the delivery to be at least the endogenous benchmark $\max_i r_i$.

Let the highest and second-highest true values be $v^{(1)} > v^{(2)}$. If the top solver reports truthfully, it sets the benchmark to $v^{(1)}$, must deliver $v^{(1)}$, and earns zero surplus. If instead it reports $r = v^{(2)} + \varepsilon$ with $0 < \varepsilon < v^{(1)} - v^{(2)}$, it still has the maximal report, the benchmark falls to $v^{(2)} + \varepsilon$, and it can earn $v^{(1)} - v^{(2)} - \varepsilon > 0$. Truthful reporting is therefore not a dominant strategy for any feasible non-wasteful value-respecting mechanism enforcing the best-bid benchmark. The same argument shows that any DSIC mechanism in this class can enforce at most the runner-up benchmark. □

Proof (Proof of Theorem 5). Lower bound. Restrict to the screening family R_g: a single strategic solver holds one batch of private value $V \in [1, g]$ (floor mass normalized to 1, so $\mathrm{FAIR}(V) = V$), and the mechanism posts an outcome as a function of the reported V. Write $p(V)$ for the probability the batch is executed and $x(V)$ for the expected batch delivery (zero when not executed). The fallback floor delivers welfare 1, so $W(V) = x(V) + 1 - p(V)$, and the solver's interim utility is $u(V) = p(V)V - x(V)$. Because bids are binding, a type V can only *under*report (claim $V' \le V$), so incentive compatibility implies

$$u(V) \ge u(V') + (V - V')p(V') \qquad (V' \le V).$$

Thus u is nondecreasing and $u'(V) \ge p(V)$ a.e. Hence $u(V) \ge \int_1^V p(t)\,dt$ and

$$x(V) = p(V)V - u(V) \le \psi(V) := p(V)V - \int_1^V p(t)\,dt.$$

Integrating $\psi(V)/V^2$ over $[1, g]$ and applying Fubini gives

$$\int_1^g \frac{\psi(V)}{V^2}\,dV = \int_1^g \frac{p(V)}{V}\,dV - \int_1^g p(t)\Big(\frac{1}{t} - \frac{1}{g}\Big)\,dt = \frac{1}{g}\int_1^g p(t)\,dt \le \frac{g-1}{g}.$$

If the mechanism has competitive ratio r, i.e. $W(V) \ge V/r$ pointwise, then $x(V) \ge V/r - (1 - p(V)) \ge V/r - 1$, and

$$\int_1^g \frac{V/r - 1}{V^2}\,dV = \frac{\ln g}{r} - \Big(1 - \frac{1}{g}\Big).$$

Combining the two inequalities, $(\ln g)/r \le 2(1-1/g)$, so every randomized DSIC mechanism has

$$r = \sup_{V\in[1,g]} \frac{V}{W(V)} \ge \frac{g\ln g}{2(g-1)} \ge \frac{\ln g}{2}.$$

Posted-price upper bound. Offer the batch a take-it-or-leave-it delivered welfare on the geometric ladder $\{e^k : 0 \le k \le K\}$, $K = \lceil \ln g \rceil$, choosing k uniformly at random; otherwise settle at floors. For any realized $V \in [e^k, e^{k+1})$, the draw k (probability $1/(K+1)$) executes the batch and delivers at least $e^k > V/e$, while all draws deliver at least the floor mass. Hence $\mathbb{E}[\text{welfare}] \ge \max\big(V/(e(K+1)), 1\big) \ge \text{FAIR}/(e(\lceil \ln g \rceil + 1))$.

Deterministic optimum. A single deterministic threshold τ executes iff $V \ge \tau$: welfare is V on executed types and the floor welfare 1 otherwise. The worst case over $V \in [1, g]$ is therefore balanced at $\tau = \sqrt{g}$, where the high-type loss g/τ and the floor loss $\tau/1$ (just below threshold) both equal $\sqrt{g}$, and no deterministic rule does better. Since R_g embeds as a degenerate single-minded subfamily of the full model, all three bounds transfer. □

Proof (Proof of Theorem 6). (i) Eligibility-posted packing. From the public floors form, for a random $k \sim \text{Unif}\{0, \dots, K\}$, $K = \lceil \ln g \rceil$, the posted weight $C_S e^k$ for each option S and admit an option iff its public value clears this weight; among admitted options take a maximum-weight disjoint packing. No reported value enters the posted weights or the packing, so the mechanism is universally truthful. Writing X for the welfare of the offline fair optimum carried by batches and $Y = C$ for the floor mass, the geometric draw clears a $1/(e(K+1))$ fraction of X in expectation while every realization secures Y, so by $\max(X/a, Y) \ge (X+Y)/(a+1)$ with $a = e(K+1)$, $\mathbb{E}[W] \ge \text{FAIR}/(e(\lceil \ln g \rceil + 1) + 1)$.

(ii) Coupled menu. Post every option at price $C_S\, t$, $t = e^k$, $k \sim \text{Unif}\{0, \dots, K\}$; the solver demands a profit-maximizing disjoint collection $\mathcal{C}^*(t)$. This is a posted-demand mechanism (the menu is independent of any report), hence universally truthful, and feasible because any demanded S has $V_S \ge C_S t \ge C_S$, so the posted delivery clears its floors. Let

$$p(t) = \max_{\mathcal{C} \text{ disjoint}} \sum_{S\in\mathcal{C}} (V_S - C_S t).$$

As a maximum of finitely many affine decreasing functions, p is convex, piecewise linear, nonincreasing, with $p(t) = 0$ for $t \ge g$ (every option has $V_S \le g C_S$). At any differentiability point $-p'(t) = \sum_{S\in\mathcal{C}^*(t)} C_S$, so the delivered mass is $D(t) = t \sum_{S\in\mathcal{C}^*(t)} C_S = t\,(-p'(t))$; at a kink the right-derivative selects the continuing (smallest-floor) active collection, and $D(t) \ge t\,(-p'_+(t))$. On $[t_k, t_{k+1}] = [e^k, e^{k+1}]$, convexity gives $-p'$ nonincreasing, so

$$\int_{t_k}^{t_{k+1}} (-p'(t))\, dt \le (-p'_+(t_k))(t_{k+1} - t_k) = (-p'_+(t_k))(e-1)e^k \le (e-1)D(t_k).$$

Summing over $k = 0, \dots, K-1$ and using $p(e^K) = 0$,

$$p(1) = p(1) - p(e^K) = \int_1^{e^K} (-p'(t))\, dt \le (e-1) \sum_{k=0}^{K} D(e^k),$$

so the uniform draw gives $\mathbb{E}[D] \ge p(1)/((e-1)(K+1))$. Since the fair optimum's batches form a feasible disjoint collection at $t = 1$, $p(1) \ge \sum_{S \in B^*}(V_S - C_S) = \mathrm{FAIR} - C$, and combining with the pointwise $W \ge C$ via the same max-inequality yields $\mathbb{E}[W] \ge \mathrm{FAIR}/((e-1)(\lceil \ln g \rceil + 1) + 1)$.

Independent pricing fails. With m options sharing one order, each priced independently on the geometric ladder, a multi-minded solver demands only the single cheapest, delivering $O(1)$ options while the fair optimum uses $\Theta(m)$; calibrating $m = \Theta(\log^2 g)$ gives the ratio $\Omega(g/\log^2 g)$. ☐

Proof (Proof of Theorem 7). Upper bound. In random-priority coupled menus, condition on a solver being served first (probability $1/N$): it faces the full coupled menu on all orders, and the profit-curve telescope of Theorem 6(ii) applies verbatim, giving expected delivery at least $p_i(1)/((e-1)(K+1))$ from that solver, where p_i is computed on the full order set. A solver served later faces a subset of orders and contributes nonnegatively. Since each solver's optimal batches form a disjoint collection on the full set, $\sum_i p_i(1) \ge \sum_{b \in B^*}(V_b - C_b) = \mathrm{FAIR} - C$. Averaging over the uniform priority and combining with $W \ge C$ gives $\mathbb{E}[W] \ge \mathrm{FAIR}/(N(e-1)(\lceil \ln g \rceil + 1) + 1)$. Truthfulness holds because a solver's menu depends only on which orders predecessors have consumed, never on its own report.

Lower bound (symmetrized swarm). Fix orders $o_1, \dots, o_N$ with floors $\varepsilon = L/(N\rho)$ and auxiliary orders $u_1, \dots, u_L$ with floor 1; let G be the set of all orders, so $C_G \le 2L$. Each identity has the *same* public option family $\{\{o_i\}, G\}$; identities differ only in private values, so any report-independent service order is statistically independent of which identity is the "batcher." Choose $\rho \in (\max\{1, g/2\}, g]$ avoiding the (countably many) atoms of the N multiplier distributions. The batcher, placed by the adversary at an identity chosen after seeing the mechanism, has $V(G) = \rho\, C_G$ and a ratio-1 singleton; every other identity (a blocker) has a ratio-ρ singleton on its own order and a ratio-1 copy of G. At multiplier t a blocker is silent iff $t > \rho$; write $q_i = \Pr[t_i > \rho]$. The batcher delivers $\Theta(\rho C_G)$ only if it is reached before any blocker has consumed an order of G, i.e. only if all predecessors are silent. Blockers together deliver at most $N\varepsilon\rho = L$, and fallback floors contribute at most $C_G \le 2L$, so pointwise $W \le \mathrm{del}_{\mathrm{batcher}} + 3L$. For any realization of the order,

$$\sum_i (1-q_i) \prod_{s \prec i} q_s \;=\; 1 - \prod_s q_s \;\le\; 1,$$

the i-th term being the probability that i is the first non-silent identity. Taking expectation over the (independent) order and the multipliers,

$$\sum_i (1-q_i) \Pr[\text{predecessors of } i \text{ silent}] \le 1,$$

so some identity i^* has $(1 - q_{i^*})\Pr[\text{predecessors silent}] \le 1/N$. Placing the batcher at i^* caps its expected batch delivery at $2\rho L/N$, while the fair optimum delivers at least ρL via the batch. Hence

$$\frac{\text{FAIR}}{\mathbb{E}[W]} \ge \frac{\rho}{3 + 2\rho/N} = \Omega(\min(N, \rho)) = \Omega(\min(N, g)),$$

using $\rho > g/2$. The blockers' geometry is public and identical across placements, so the bound holds even for geometry-aware orders. □

D Full Proofs for Section 6

Proof (Proof of Proposition 3). Settle-at-deadline is fair and achieves C on every instance, while FAIR $\le$ OPT $\le gC$, so the ratio is at most g; a staggered-deadline instance forcing individual-only settlement shows it is at least g. □

Proof (Proof of Theorem 8). (i) If an almost-surely fair mechanism settles an exposed batch (a covered order's deadline in the future) with positive probability on some prefix, fix that prefix and append a higher standalone bid on a later-deadline covered order: on the appended instance it has, with positive probability, settled below a realized floor—not a.s. fair. Hence exposed batches are never settled, and on the staggered hard family the mechanism is individual-only a.s., with welfare $\le C$ against FAIR $= gC$. (ii) Stopped welfare $q = V + C - C_S$ satisfies $q \in [C, gC]$ ($V \ge C_S$ and $V \le gC_S$, $C_S \le C$). *Deterministic* $\Theta(\sqrt{g})$: accept the first $q \ge \sqrt{g}\,C$; if $q^* \ge \sqrt{g}\,C$ the ratio is $q^*/q \le \sqrt{g}$, else $W = C$ and ratio $< \sqrt{g}$; a two-step adversary matches. *Randomized* $O(\log g)$: with $k \sim \text{Unif}\{0, \dots, K\}$, accept the first $q \ge e^k C$. With probability $1/(K+1)$, $e^k C \le q^* < e^{k+1} C$, and then the mechanism accepts some offer with $q \ge e^k C > q^*/e$, so $\mathbb{E}[W] \ge q^*/(e(K+1))$. The matching $\Omega(\log g)$ is a Yao bound: a persistent order with fleeting geometric batch offers $x_i = e^i c$, $i \le \lfloor \ln g \rfloor$, shown for the first M with $\Pr[M \ge i] = e^{-i}$, caps every deterministic threshold at $2c$ while $\mathbb{E}[\text{FAIR}] = \Omega(c \log g)$. Greedy: a ratio-$g$ blocker arriving before a ratio-g batch is accepted (killing the batch) for any threshold $T \le g$ and rejected with the batch for $T > g$, ratio $\Omega(g)$. □

Proof (Proof of Theorem 9). In the per-order-decomposable individual game, the maximal standing bid on j both wins and delivers itself, so $c_j(b) = p_j(b)$. Let $i^*(j)$ deviate to W_j with density $1/(v_j^{\max} - w)$ on $[0, (1 - 1/e)v_j^{\max}]$ (Syrgkanis–Tardos). Then the expected utility from j is $((1 - \frac{1}{e})v_j^{\max} - \beta_j)^+ \ge (1 - \frac{1}{e})v_j^{\max} - p_j(b)$ with $\beta_j \le c_j(b) = p_j(b)$, the penalty landing under b. Summing and using $\sum_i u_i(b) = \text{PW}(b) - \sum_j p_j(b)$ in the CCE inequality cancels the price terms, giving $\mathbb{E}[\text{PW}] \ge (1 - 1/e)\text{OPT}_{\text{ind}}$; against the unrestricted optimum, OPT $\le g\,\text{OPT}_{\text{ind}}$, so the factor is at most $(e/(e-1))g$. The pure-NE lower bound is Theorem 1's instance with a filtered $(2gv, 0)$ batch. The full-game obstruction is a latent batch present in b_{-i} that executes under the deviation but not under b, breaking every charging scheme. □

Proof (Proof of Theorem 10). The steady-state benchmark under a constant injection a is $b^* - \beta a$ (the rate limit does not bind once the bias is constant), so the per-round profit is $\pi(a) = v\beta a - \kappa_1 a - \frac{\kappa_2}{2}a^2$. With $\kappa_2 > 0$, π is strictly concave with $\pi'(0) = v\beta - \kappa_1$; hence if $v\beta \le \kappa_1$ the unique maximizer is $a^* = 0$ (no attack), and otherwise $a^* = \min\{\bar{a}, (v\beta - \kappa_1)/\kappa_2\}$, with relative bias $\mu = \beta a^*/b^*$.

Bursts do not help. The published benchmark satisfies $b_t \ge b^* - \beta a_t$ pointwise, so for every horizon T and every (possibly time-varying) policy (a_t),

$$\sum_{t=1}^{T} \big[v(b^* - b_t) - K(a_t)\big] \le \sum_{t=1}^{T} \big[v\beta a_t - K(a_t)\big] = \sum_{t=1}^{T} \pi(a_t) \le T\,\pi(a^*),$$

and the constant policy a^* attains $T\pi(a^*) - O(1)$ after the ramp; hence no episodic schedule beats the constant attack.

Ramp tax (exact). Starting from $b_0 = b^*$, the rate limit caps the attainable bias at round t by $\Delta_t := \min\{\Delta^*,\, b^*(1-(1-r)^t)\}$ with $\Delta^* = \beta a^*$; reaching Δ^* takes $T_\mu = \left\lceil \ln \frac{1}{1-\mu} \big/ \ln \frac{1}{1-r} \right\rceil$ rounds. An attacker who hugs the cap injects $a_t = \Delta_t/\beta$, so the exact net rent forgone per restart is

$$L_{\mathrm{net}} = \sum_{t=1}^{T_\mu} \Big[\pi(a^*) - \pi(\Delta_t/\beta)\Big] \;\le\; \pi(a^*)\,T_\mu.$$

In the small-(r,μ) regime, $1-(1-r)^t \approx rt$ and the summand is $\frac{\kappa_2}{2}(a^*)^2(1-rt/\mu)^2$, giving $L_{\mathrm{net}} = \frac{\pi(a^*)\mu}{3r}(1+o(1))$; we state this only as a first-order approximation, the exact figure being the finite sum above.

Calibration. A steady bias μ scales every floor on the pair by $1-\mu$, so any guarantee whose only calibration input is the public-floor spread $G = \max_i V_i/C_{S_i}$ becomes $G/(1-\mu)$; the coverage law recalibrates jointly, $(g,\alpha) \mapsto (g/(1-\mu), \min\{1, \alpha/(1-\mu)\})$. Structural production complementarity, directed-pair complexity, and produced-welfare PoA are unaffected. □

E Deployment Diagnostics

Table 2. Deployment diagnostics used only for calibration. The visible spread is auction-level and observable; it is not the structural complementarity factor g.

Diagnostic	Value or interpretation
Competitions in trace	802,816 post-adoption solver competitions
Covered notional / gap	\$54.69B / \$5.13M (about 0.94 bps)
Visible spread tail	$\widehat{G}^{\mathrm{obs}} > 2$: 25.3%; $\widehat{G}^{\mathrm{obs}} > 4$: 20.2%
Small-notional tail	$\widehat{G}^{\mathrm{obs}} > 2$: 63.1%
Directed-pair support P	directly observable from requested/paid pairs
Naive local reserve	97.1% vulnerable cell-windows in diagnostic